EXPERIMENT 7

Comparative Study of Popular No-Code / Low-Code Application Development Platforms

Subject: No-Code Application Development

Aim

To compare widely used no-code and low-code platforms for building business applications with minimal conventional programming.

Objectives

- Identify core features and intended users of different platforms.
- Compare integrations, customization, data management and deployment.
- Recommend a platform for a sample business problem.

Selected Application

A small employee leave request and tracking system requiring a form, centralized records, a manager approval stage and a status dashboard.

Requirements / Tools

- Product documentation and publicly available platform demos
- Evaluation criteria: ease of use, workflow support, integration and pricing
- Candidate platforms: Zoho Creator, Microsoft Power Apps, Google AppSheet and Bubble

No-Code Connection

Visual application development uses drag-and-drop interfaces, reusable components and workflow rules. Low-code additionally allows expressions, APIs or scripts when visual tools are insufficient.

Illustrative Business Scenario

Compare each platform against the same requirement: a leave-request form, manager approval, dashboard and optional email alert.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Define the sample application features and likely number of users.
2. Review each platform’s form builder, data sources, workflow automation and publishing process.
3. Note platform-specific scripting or formula options and external integrations.
4. Compare collaboration, mobile access, permissions, data portability and pricing terms.
5. Score the platforms qualitatively for the sample use case and explain the recommendation.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

Platform comparison

Platform	Strength	Consideration
Zoho Creator	Forms, reports, Deluge workflows	Zoho-specific tooling and licensing
Microsoft Power Apps	Microsoft 365 and Dataverse integration	Licensing and platform complexity
Google AppSheet	Spreadsheet-driven mobile/web apps	Complex UI control can be limited
Bubble	Flexible visual web apps	Learning curve and platform dependence

Evaluation matrix (qualitative)

Criterion	Zoho Creator	Power Apps	AppSheet	Bubble
Business forms	Strong	Strong	Strong	Good
Approvals/workflows	Strong	Strong	Good	Custom setup
Rapid data app	Strong	Strong	Strong	Good
Custom web UI	Moderate	Moderate	Moderate	Strong

Interpretation / Analysis

- No-code platforms reduce setup and coding effort for standardized workflows, but advanced requirements can still require configuration or scripting.
- Enterprise integration, data ownership, user licensing and vendor lock-in should be evaluated before adoption.
- The qualitative ratings describe typical platform positioning rather than tested benchmark scores.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

For the selected approval-based internal business application, Zoho Creator is a suitable candidate because forms, reports and workflow rules can be developed in one environment; existing Microsoft or Google ecosystems may favor alternatives.

Conclusion

Platform selection should be driven by business requirements and integration needs, not only visual design or development speed.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.